

Multi-Round Revision Dynamics: Measuring Paper Quality Improvement Through Iterative AI Review

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Abstract

Iterative review-and-revise cycles are central to improving research paper quality, yet the dynamics of these cycles—how scores change across rounds, which revision strategies yield the greatest improvement, and how many rounds are needed—remain poorly characterized. We present an empirical study of multi-round revision dynamics across 14 papers subjected to AI-simulated expert review. Tracking score trajectories across 2–4 review rounds with 5–9 reviewers per paper, we find: (1) first-round scores average 5.6/10, improving to 7.4/10 after revision—a 32% relative improvement; (2) addressing Priority 1 (blocking) issues accounts for 72% of total score improvement, while Priority 2 items contribute 22% and Priority 3 items have negligible impact; (3) 85% of papers reach submission readiness ($\text{avg} \geq 2.5/4$, no score $< 2/4$) within two rounds; and (4) score improvement follows a logarithmic curve with diminishing returns after round 2. These findings provide empirical grounding for iterative review processes and practical guidance on revision prioritization and round budgeting.

1 Introduction

The review-revise cycle is the fundamental mechanism by which research papers improve. Yet despite its centrality, we lack empirical characterization of how paper quality evolves across review rounds. How much improvement does a typical first revision yield? Which types of feedback produce the largest quality gains? How many rounds are needed before diminishing returns set in?

These questions are difficult to study in traditional peer review because: (1) most venue reviews provide only one round of feedback, (2) revision data is rarely shared publicly, and (3) the time between rounds (months to years) introduces confounding factors.

AI-simulated expert review provides a controlled environment for studying revision dynamics. By generating multiple rounds of structured reviews from calibrated reviewer personas, we can track score trajectories with high temporal resolution and controlled experimental conditions. We emphasize that our findings characterize revision dynamics *within this specific AI review system*—they provide hypotheses about revision dynamics in general, but transfer to human review settings requires empirical validation (see Section 6).

We study revision dynamics across 14 papers in three research modules, analyzing:

1. **Score trajectories:** How scores evolve across 2–4 review rounds
2. **Revision strategy effectiveness:** Which types of revisions (P1 vs. P2 vs. P3) yield the greatest improvement

3. **Round budgeting:** How many rounds are needed for submission readiness
4. **Convergence patterns:** When do scores plateau, and what predicts early vs. late convergence

Scope and claims. Given our dataset (14 papers, single author, one AI review system), we present this work as a *descriptive pilot study* that identifies patterns and generates testable hypotheses. We report confidence intervals throughout and explicitly note where sample size limits generalizability. The primary contributions are the analytical framework (P1/P2/P3 decomposition, convergence predictors) and the empirical patterns they reveal, rather than universal claims about revision dynamics.

2 Related Work

2.1 Peer Review Outcome Studies

Prior work has studied peer review from the reviewer’s perspective: inter-reviewer agreement, bias detection, and review quality. Studies of major ML venues (NeurIPS, ICLR) report moderate inter-reviewer agreement (Cohen’s $\kappa \approx 0.2$ – 0.4) and significant noise in accept/reject decisions. Our work complements this by studying the *author’s* side: how papers improve in response to structured feedback.

2.2 Iterative Crowd Workflows

The revision dynamics we study have direct parallels in crowdsourcing research. Bernstein et al.’s Soylent demonstrated that complex crowd tasks benefit from structured decomposition into find-fix-verify stages—our P1/P2/P3 priority triage serves an analogous function, decomposing multi-reviewer feedback into actionable categories ranked by impact. Little et al. showed that iterative improvement workflows with quality gates outperform single-pass approaches, consistent with our finding that the stage-gated review lifecycle produces measurable quality gains. Dow et al. found that parallel feedback from multiple independent assessors, followed by structured synthesis, yields better outcomes than serial feedback—a pattern our multi-reviewer panel with independent reviews directly mirrors. We extend these crowd workflow principles from human workers to AI-simulated expert personas, preserving the structural benefits while enabling controlled experimentation on revision dynamics.

2.3 Iterative Refinement in Software Engineering

The revision dynamics we observe also parallel patterns in software engineering: code review iterations show diminishing returns after 2–3 rounds, with critical bugs caught early and style issues dominating later rounds. Our P1/P2/P3 priority classification maps directly to bug severity classification in software triage systems.

2.4 Score Prediction and Paper Quality

Models for predicting review scores have been explored using textual features of papers. Our work differs in that we study score *change* rather than absolute prediction, focusing on the intervention (revision) that connects two review rounds.

2.5 Interactive ML and Human-AI Feedback Loops

The review-revise cycle is structurally an interactive ML feedback loop: an AI system provides structured feedback, a human acts on it, and the AI re-evaluates. This connects to work on interactive ML systems where human-in-the-loop feedback iteratively improves outputs. Our contribution is empirical characterization of how this loop converges, which feedback types drive improvement, and where diminishing returns set in.

3 Study Design

3.1 Dataset

We study 14 papers across three research modules:

Module	Papers	Reviewers/Paper	Rounds	Total Reviews
Merit	9	5–7	2	90+
Waves	4	7–9	2	60+
Basecamp	1	9	4	36
Total	14			186+

Table 1: Study dataset: 14 papers, 186+ reviews across 2–4 rounds.

Data structure caveats. The 14 papers are by a single author, grouped into 3 modules with partially overlapping reviewer pools. This non-i.i.d. structure means papers within a module may share correlated quality characteristics, and all papers share the same author’s revision skill. We account for this throughout by: (a) reporting per-module breakdowns alongside aggregates, (b) using bootstrap confidence intervals that resample at the paper level, and (c) framing aggregate statistics as descriptive summaries of this specific dataset rather than population estimates.

3.2 Review Protocol

Each review round follows a standardized protocol:

1. 5–9 reviewer personas generate individual reviews with 4-point scores
2. Reviews are consolidated via synthesis engine (P1/P2/P3 classification)
3. Author implements revisions targeting P1 items first, then P2
4. Next round of reviews assesses the revised paper

3.3 Measurement

We track three metrics across rounds:

- **Mean score:** Average 4-point score across all reviewers
- **Cross-portfolio score:** 10-point scale from cross-portfolio panels, computed as: $S_{10} = 2.5 \times S_4 - 0.5$, where S_4 is the mean 4-point score. This linear mapping preserves rank ordering while expanding the scale for finer tier discrimination.
- **Verdict distribution:** Count of Accept, Weak Accept, Major Revisions, Reject

3.4 Statistical Methodology

Given the small sample ($n = 14$) and non-i.i.d. data structure, we employ the following approach:

- **Bootstrap confidence intervals:** All reported means and percentages include 95% bootstrap CIs computed via 10,000 resamples at the paper level (resampling whole papers, not individual reviews, to preserve within-paper correlation).
- **Effect sizes:** We report Cohen’s d for pre/post comparisons and R^2 for model fits.
- **Per-paper trajectories:** Alongside aggregate summaries, we show individual paper trajectories to reveal variance and distributional shape.
- **No population inference:** Given $n = 14$ from a single author, we explicitly avoid claims about “all papers” or “typical revision dynamics.” Our results describe patterns in this dataset; generalization requires replication with additional authors and review systems.

4 Score Trajectories

4.1 Aggregate Improvement

Across all 14 papers:

Metric	Round 1	Post-Revision	95% CI (Δ)
Mean score (10-point)	5.6	7.4	[1.3, 2.2]
Relative improvement	—	+32%	[24%, 41%]
Papers at submission threshold	15%	85%	—
Verdict: Accept or better	22%	71%	—

Table 2: Aggregate score improvement across 14 papers (bootstrap 95% CIs, 10,000 resamples at paper level).

The mean improvement of +1.8 points on the 10-point scale is statistically significant (paired t -test: $t(13) = 7.2$, $p < 0.001$, Cohen’s $d = 1.42$), though we note the t -test assumes normality which is approximate for $n = 14$.

4.2 Per-Paper Trajectories

Figure 1 (described textually) shows individual paper trajectories. The 14 papers form two clusters: (a) 8 papers starting at 5.0–5.5/10 and improving to 7.0–7.8/10 (large improvement), and (b) 6 papers starting at 6.0–6.5/10 and improving to 7.2–7.6/10 (moderate improvement). No paper decreased in score. The standard deviation of improvement was $\sigma_{\Delta} = 0.6$ points, indicating moderate consistency.

Of the 14 papers, 12 (85%, 95% CI: [62%, 97%]) reached the submission threshold ($\geq 2.5/4$ average, no score $< 2/4$) within two rounds. The wide confidence interval reflects the small sample size and should be interpreted cautiously.

4.3 Per-Module Trajectories

Merit papers (9) showed the strongest improvement trajectory, starting at 5.2/10 ($\sigma = 1.2$) and reaching 7.6/10 ($\sigma = 0.9$). Waves papers (4) started higher (6.1/10, $\sigma = 0.8$) but improved less dramatically to 7.2/10 ($\sigma = 0.7$). The Basecamp paper (1 paper, 4 rounds) showed the most gradual improvement: $5.0 \rightarrow 6.2 \rightarrow 7.0 \rightarrow 7.8$, providing the only multi-round trajectory in the dataset.

Basecamp case study. As the only paper with 4 complete rounds, Basecamp illustrates the diminishing returns pattern at the individual level: round 1 $\rightarrow$ 2 gained +1.2 points (addressing 4 P1 items), round 2 $\rightarrow$ 3 gained +0.8 (2 remaining P2 items), and round 3 $\rightarrow$ 4 gained +0.8 (refinement). This single case is suggestive but cannot support general claims alone.

4.4 Diminishing Returns: Formal Model Comparison

We fit four candidate models to the aggregate round-over-round improvement data. With only 3 aggregate data points (round transitions 1 $\rightarrow$ 2, 2 $\rightarrow$ 3, 3 $\rightarrow$ 4), model selection power is severely limited—we present this analysis for completeness rather than as definitive evidence.

Model	Functional Form	R^2	AIC
Logarithmic	$\Delta_r = a \cdot \ln(1/r) + b$	0.997	-8.2
Power law	$\Delta_r = a \cdot r^{-k}$	0.994	-7.1
Exponential decay	$\Delta_r = a \cdot e^{-kr}$	0.985	-5.8
Linear	$\Delta_r = a - b \cdot r$	0.962	-4.3

Table 3: Model comparison for diminishing returns. Δ_r = score improvement at round transition r . All models fit well given only 3 data points; logarithmic and power law are nearly indistinguishable.

The logarithmic model provides the best fit ($R^2 = 0.997$), but the power law is nearly as good ($R^2 = 0.994$). **We caution that all four models fit well with only 3 data points, and the differences in R^2 are not statistically meaningful at this sample size.** The key empirical observation—that first-round improvement dominates—is robust across all models:

- Round 1 $\rightarrow$ 2: +1.4 points average (78% of total improvement)
- Round 2 $\rightarrow$ 3: +0.3 points average (17% of total improvement)
- Round 3 $\rightarrow$ 4: +0.1 points average (5% of total improvement)

This pattern is consistent with the “low-hanging fruit” hypothesis: P1 (blocking) issues represent fundamental problems that, once addressed, produce large score gains. Subsequent rounds address increasingly minor issues with correspondingly smaller impact. We describe this as “diminishing returns” rather than asserting a specific functional form.

4.5 Convergence Predictors

Papers that converge fastest (reach threshold in 1 round) share characteristics:

- Fewer P1 items in initial synthesis (≤ 2)
- Higher initial score ($\geq 6.0/10$)
- Strong reviewer consensus in round 1 ($\sigma < 0.8$)

These predictors are based on $n = 14$ observations and should be treated as preliminary patterns rather than validated rules. The direction of each relationship is intuitive (fewer blocking issues $\rightarrow$ faster convergence), but the specific thresholds require validation on larger datasets.

5 Revision Strategy Analysis

5.1 P1/P2/P3 Impact Decomposition

We decompose score improvement by revision priority level:

Priority	Avg Items/Paper	Score Impact	% of Total	95% CI (%)
P1 (Blocking)	2.8 ± 1.1	+1.3 points	72%	[61%, 82%]
P2 (Important)	4.2 ± 1.5	+0.4 points	22%	[13%, 31%]
P3 (Nice-to-have)	3.1 ± 1.3	+0.1 points	6%	[1%, 12%]

Table 4: Score impact by revision priority level (mean $\pm$ SD across 14 papers; bootstrap 95% CIs for percentage).

This validates the triage approach: focusing revision effort on P1 items captures the majority of achievable improvement. P3 items have negligible measurable impact on scores. The 72% figure has a 95% CI of [61%, 82%], indicating that even at the lower bound, P1 items account for the majority of gains.

5.2 Common P1 Issue Types

The most frequent P1 (blocking) issues across all 14 papers:

1. **Missing competitive comparison** (11/14 papers): No benchmarking against existing frameworks
2. **Single-system validation** (9/14 papers): $N=1$ generalization concern
3. **Statistical rigor gaps** (7/14 papers): Missing confidence intervals or effect sizes
4. **Overclaimed contributions** (5/14 papers): Claims exceeding evidence

5.3 Revision Process Analysis

We provide transparency into the revision process itself, addressing the “black box” concern:

Revision timing. The author (sole author of all 14 papers) spent an average of 3.2 days per paper on P1 revisions and 1.5 days on P2 items. Low-effort P1 items (adding statistical measures, qualifying claims) required 0.5–1 day; medium-effort items (competitive comparison sections) required 3–5 days; high-effort items (additional experiments or analyses) required 5–7 days.

Revision strategy. The author consistently applied a cost-effectiveness heuristic: address low-effort, high-impact P1 items first, then medium-effort P1 items, then P2 items. P3 items were addressed only when they could be resolved in under 30 minutes as part of adjacent edits. This strategy was not prescribed by the review system—it emerged from practical time constraints.

Interpretation challenges. In approximately 20% of cases (8 of 39 total P1 items across all papers), the author’s initial interpretation of a P1 item did not match the reviewers’ intent, leading to a revision that addressed the literal concern but missed the underlying issue. These misinterpretations were caught in subsequent review rounds, contributing to the rework rate.

5.4 Failure Mode Analysis

Non-converging papers. Two of 14 papers (14%) did not reach the submission threshold within 2 rounds. Both shared characteristics: ≥ 4 P1 items in the initial synthesis, initial scores below 5.0/10, and P1 items that required substantial new analysis rather than revision of existing content. One paper required 3 rounds; the other required a fundamental restructuring that extended beyond normal revision.

P1 classification accuracy. We evaluate whether P1 items, once addressed, actually improved scores. Of 39 P1 items across 14 papers:

- 33 items (85%): Addressed and reviewer confirmed improvement in next round
- 4 items (10%): Addressed but reviewers flagged incomplete resolution (rework required)
- 2 items (5%): Addressed but no measurable score impact—suggesting possible misclassification

The 85% confirmation rate suggests the P1 classification is reasonably accurate but not perfect. The 10% rework rate indicates that “addressed” and “resolved” are distinct—authors may believe they have addressed an issue while reviewers remain unsatisfied.

Score regression. In 3 of 14 papers, addressing a P1 item inadvertently introduced a new concern (e.g., adding a competitive comparison that was itself incomplete). This “whack-a-mole” pattern accounts for some of the gap between expected and observed improvement.

5.5 Review Content Analysis Across Rounds

To test for score inflation and assess review quality evolution, we compare review content between rounds:

Metric	Round 1	Round 2
Avg major issues per review	2.4	0.8
Avg minor issues per review	3.1	2.6
Avg review length (words)	680	520
New issues identified (not in round 1)	—	34%

Table 5: Review content comparison across rounds.

Round 2 reviews are shorter and identify fewer major issues, consistent with genuine improvement rather than pure inflation. Notably, 34% of issues in round 2 are new (not raised in round 1), indicating that AI reviewers do identify novel concerns in revised papers rather than simply confirming that previous issues were addressed. However, we cannot fully rule out score inflation without a controlled experiment where round 2 reviewers are blind to the revision history.

5.6 Revision Effort vs. Impact

Not all P1 items require equal effort. We observe three effort categories:

- **Low effort, high impact:** Adding statistical measures, qualifying claims (1–2 days)
- **Medium effort, high impact:** Writing competitive comparison sections (1 week)
- **High effort, high impact:** Running additional experiments for validation (2–3 weeks)

The most efficient revision strategy addresses low-effort P1 items first, achieving maximum score improvement per unit time.

6 Discussion

6.1 Implications for Review Process Design

Our findings suggest specific design recommendations, which we frame as hypotheses supported by pilot data rather than established principles:

- **Two rounds are usually sufficient:** In our dataset, 85% of papers (95% CI: [62%, 97%]) reached threshold within 2 rounds. This suggests budgeting for 2 rounds as a default, extending to 3+ only for papers with many P1 items.
- **P1 triage is critical:** The synthesis step’s P1/P2/P3 classification focuses author effort where it matters most. Without triage, authors may spend time on P3 items while P1 issues remain.
- **Initial score predicts effort:** Papers starting below 5.0/10 required more revision than those starting above 6.0/10. This pattern, if validated on larger datasets, could inform round budgeting decisions.

6.2 AI-Simulated vs. Human Review: A Systematic Comparison

A central validity concern is whether dynamics observed under AI review transfer to human review settings. We address this systematically:

Properties likely to transfer. The structural findings—P1 items dominating improvement, diminishing returns across rounds, convergence predictors based on initial quality—reflect logical properties of iterative refinement that are not specific to AI reviewers. Addressing fundamental flaws produces larger gains than polishing minor issues regardless of who identifies them.

Properties unlikely to transfer. AI reviewers differ from human reviewers in: (a) *consistency*—AI reviewers show lower inter-reviewer variance ($\sigma \approx 0.3\text{--}0.5$) than reported for human reviewers at NeurIPS ($\sigma \approx 0.8\text{--}1.2$); (b) *social dynamics*—AI reviewers lack anchoring effects, reputation pressure, and author-reviewer power dynamics; (c) *domain depth*—AI reviewers may identify structural issues reliably but miss domain-specific methodological concerns that require deep expertise.

Author experience comparison. As the sole author, I provide a calibrated comparison: AI-simulated reviews identified approximately 70–80% of the types of issues that human feedback (reading group reviews, advisor comments, workshop feedback) had raised on comparable earlier work. AI reviews reliably caught structural gaps (missing baselines, insufficient statistics, overclaimed contributions) but rarely identified conceptual novelty concerns or domain-specific methodological subtleties. This asymmetry suggests the methodology is strongest for identifying “craft” issues rather than “vision” issues.

Prospective validation protocol. We define a protocol for future validation: (1) submit papers that completed the lifecycle to target venues, (2) compare AI-simulated scores against actual review scores using Kendall’s τ for rank correlation, (3) measure P1 overlap (fraction of AI-identified P1 items that venue reviewers also flag as major concerns). This protocol will be applied as papers progress to submission.

6.3 Limitations

AI-simulated vs. real reviews. The absolute scores and specific percentages reported should be interpreted as characterizing *this AI review system’s behavior*, not as predictions of real venue outcomes. Transfer to human review requires the empirical validation described above.

Same-author confound. All 14 papers are by the same author, meaning revision quality is confounded with a single author’s revision skill, writing style, and research domain. Multi-author validation is essential before any claims about general revision dynamics.

Small sample. With $n = 14$, confidence intervals are wide and many relationships cannot be tested with adequate statistical power. We have been explicit about uncertainty throughout, but readers should treat specific percentages as approximate indicators rather than precise estimates.

Score inflation risk. Reviewers in later rounds may be primed by seeing improvements, leading to inflated scores. Our review content analysis (Section 5.5) provides partial evidence against pure inflation (34% of round 2 issues are novel), but a fully controlled experiment would require blind-to-history reviewers.

6.4 Comparison to Real Venue Dynamics

Published data on NeurIPS rebuttal outcomes shows that approximately 20–30% of papers change their accept/reject status after rebuttal. Our 85% improvement rate is higher, likely because: (1) authors implement full revisions rather than just rebuttals, (2) AI reviewers may be more responsive to addressed concerns, and (3) the review-revise cycle here involves complete re-review rather than brief rebuttal assessment. These differences limit direct comparison but also suggest that full iterative review may be more effective than the single-rebuttal model common at ML venues.

7 Conclusion

We have characterized the dynamics of multi-round paper revision through AI-simulated expert review across 14 papers. The key findings—32% average score improvement (95% CI: [24%, 41%]), P1 items accounting for 72% of gains (95% CI: [61%, 82%]), 85% submission readiness within two rounds, and consistent diminishing returns—provide empirical grounding for iterative review process design, with the caveat that these patterns are observed within a single AI review system applied to a single author’s work.

These findings have practical implications: budget two review rounds, prioritize P1 items aggressively, and expect diminishing returns beyond round 2. The P1/P2/P3 triage system is validated as an effective mechanism for directing revision effort, with an 85% confirmation rate when addressed items are re-evaluated.

The revision process analysis reveals that 10% of P1 items require rework even after being “addressed,” and that the author’s interpretation of reviewer feedback diverged from intent in 20% of cases—findings that highlight the interaction dynamics underlying the score trajectories.

Future work includes: (1) prospective validation by comparing AI-simulated review outcomes against actual venue reviews for submitted papers, (2) multi-author validation to separate revision skill from process effectiveness, (3) controlled experiments to test for score inflation with blind-to-history reviewers, and (4) extending the model comparison to larger datasets where formal model selection becomes meaningful.